

Introduction:

HTTP(Hyper Text Transfer protocol):

is an application-layer protocol used to exchange web requests and responses between a client and a server. A standard HTTP request uses TCP port 80 and does not provide encryption, which means the transmitted data can be read by anyone who intercepts the traffic.

HTTPS URL Breakdown:

```
http://cybersecurity.com
```

Component Description

http Application-Layer Protocol

:// Separator

cybersecurity.com Domain name

we can send get request on the by using the http get request to access to any web page like as i send this request <http://cybersecurity.com>

HTTP GET Request Flow (Client Side) Encapsulation:

User types:

<https://google.com>

|

Browser

|

DNS Resolution

|

IP Address Found

|

Application Layer (HTTP)

creates HTTP Get Request

|

Transport layer (TCP)

Adds TCP Header

```
Creates TCP Segments
|
Network Layer (IP)
Creates IP Packet
|

Data Link Layer (Ethernet)
Adds Ethernet Header
Creates Ethernet Frame
|
Physical Layer
converts bits into electrical /
optical / wireless signals
|
Internet / ISP / Routers
|
Destination Server
```

HTTP GET Request Flow (Server Side) Decapsulation:

```
Physical Layer
Receives Signals
|
Data Link Layer
Removes Ethernet Header
|
Network Layer
Removes IP header
|
Transport Layer
Removes TCP Header
Reassembles Segments
|
Application Layer
Process HTTP get request
|
Web server
Finds Requested page
|
Creates HTTP Response
```

Response Back to Client:

```
HTTP Response
|
Application Layer
|
Transport Layer (TCP)
|
Network Layer (IP)
|
Data Link Layer(Ethernet)
|
Physical Layer
|
Internet
|
client Physical Layer
|
Data Link Layer
|
Network layer
|
Transport layer
|
Application Layer
|
Browser Displays Web Page
```

DNS Resolution

the browser first performs DNS resolution to obtain the server's IP address

```
User
|
Browser
|
DNS Cache
|
Operating System Cache
|
Router Cache
|
Recursive DNS Resolver
```

```
|  
Root DNS Server  
|TLD Server(.com)  
|  
Authoritative DNS Server  
|  
IP Address
```

the browser accepts a domain name from the user, but it must perform DNS resolution to obtain the server's IP address before establishing a connection of the server it needs IP address so it goes to the DNS server that checks whether it is .com .org or anything else as Google has .com so it goes to .com server and then the DNS server gave the IP address sometime it is saved in the DNS cache so that when we search about the same domain then it will get the IP address directly from that it saves time after getting the domain name .

Encapsulation:

flow chart:

```
Application Layer  
|  
Transport Layer (TCP)  
|  
Network Layer (IP)  
|  
Data Link Layer (Ethernet)  
|  
Physical Layer
```

Application Layer:

is an application-layer protocol used to exchange web requests and responses between a client and a server. A standard HTTP request uses TCP port 80 and does not provide encryption, which means the transmitted data can be read by anyone who intercepts the traffic. When a user enters a URL such as <http://cybersecurity.com> the browser first performs DNS resolution to obtain the server's IP address. The application layer creates the HTTP request as a sequence of bytes.

Transport Layer:

The data from the application layer is passed to the transport layer. The transport layer used to TCP for standard HTTP and HTTPS communication and address a TCP header containing information such as source port, destination port, sequence number, and acknowledgment number.

Network Layer:

transport layer send the segments to the network layer. network layer is used to select the best path to send that now in the network layer these segments are converted to the packets network layer add the its header that contain ip address(The Ip header contain the source and the destination ip address which allows routers to send the packets by different paths but correct destination) network layer use the Router that is layer three device use the ip addresses to select the best path to send the packet.Packets may take different paths through the network. TCP is responsible for reordering segments and delivering the data to the application in the correct sequence. these all packets are send in the form of sequences then it send it to the data link layer.

Data Link Layer:

data link layer used physical address that we call mac addresses. These are used for local communication in the same network . The Data Link Layer encapsulates the network-layer packet inside a frame.

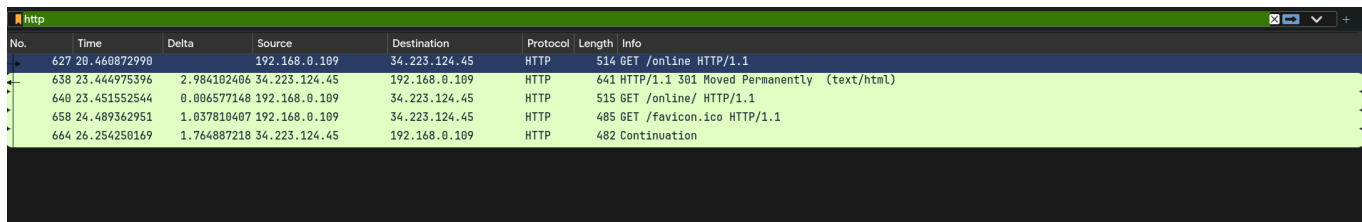
The network layer then encapsulates the TCP segment into an IP packet by adding the source and destination IP addresses. The Data Link Layer encapsulates the packet into an Ethernet frame by adding source and destination IP addresses. The Data Link Layer encapsulates the IP packet into Ethernet frame by adding source and destination MAC addresses.

Physical Layer:

Finally, the physical layer transmits the frame as electrical,optical,or wireless signals through the network. At the destination, the process is reversed through decapsulation until the application layer receives the original HTTP request.

Wireshark (capturing network traffic):

Packet List:

A screenshot of the Wireshark Packet List pane. The title bar says 'http'. The table has columns: No., Time, Delta, Source, Destination, Protocol, Length, and Info. The first packet (No. 627) is a GET request for /online. The second packet (No. 638) is a 301 Moved Permanently response. The third packet (No. 640) is a GET request for /online/. The fourth packet (No. 658) is a GET request for /favicon.ico. The fifth packet (No. 664) is a Continuation response.

No.	Time	Delta	Source	Destination	Protocol	Length	Info
627	20.468872998		192.168.0.109	34.223.124.45	HTTP	514	GET /online HTTP/1.1
638	23.444975396	2.984102406	34.223.124.45	192.168.0.109	HTTP	641	HTTP/1.1 301 Moved Permanently (text/html)
640	23.451552544	0.006577148	192.168.0.109	34.223.124.45	HTTP	515	GET /online/ HTTP/1.1
658	24.489362951	1.037810407	192.168.0.109	34.223.124.45	HTTP	485	GET /favicon.ico HTTP/1.1
664	26.254250169	1.764887218	34.223.124.45	192.168.0.109	HTTP	482	Continuation

Captured packets:

- 627 GET /online HTTP/1.1
 - 638 HTTP/1/1 301 Moved Permanently
 - 640 GET /online HTTP/1.1
 - 658 GET /favicon.ico HTTP/1.1
 - 664 Continuation
-

Wireshark Packet Overview Of Frame 627:

The captured packet (Frame 627) represents an HTTP GET request sent from my client machine (192.168.0.109) to the web server (34.223.124.45). Wireshark shows that this packet is encapsulated through multiple protocol layers. Starting from the application layer (HTTP), the data is encapsulated inside a TCP segment, then inside an IPv4 packet, and finally inside an Ethernet frame before being transmitted over the physical network interface (wlan0).

The packet contains the following protocol layers:

- Frame - General information about the captured packet.
 - Ethernet II - Data Link Layer information containing source and destination MAC addresses.
 - Internet Protocol Version 4 (IPv4) - Network Layer information containing the source and destination IP addresses.
 - Transmission Control Protocol (TCP) - Transport Layer information responsible for reliable communication.
 - Hypertext Transfer Protocol(HTTP)- Application Layer protocol carrying the HTTP GET request.
-

EthernetII Header Analysis:

```

> Frame 627: Packet, 514 bytes on wire (4112 bits), 514 bytes captured (4112 bits) on interface wlan0,
> Ethernet II, Src: LiteonTechno_31:3f:09 (24:b2:b9:31:3f:09), Dst: TendaTechnoL_64:9d:90 (b4:0f:3b:64:9d:90)
  > Destination: TendaTechnoL_64:9d:90 (b4:0f:3b:64:9d:90)
  > Source: LiteonTechno_31:3f:09 (24:b2:b9:31:3f:09)
  > Type: IPv4 (0x0800)
  [Stream index: 0]
> Internet Protocol Version 4, Src: 192.168.0.109, Dst: 34.223.124.45
> Transmission Control Protocol, Src Port: 50032, Dst Port: 80, Seq: 1, Ack: 1, Len: 448
> Hypertext Transfer Protocol

```

The Ethernet II header is added by the Data Link Layer before the packet is transmitted on to the local network. It contains the source and destination MAC addresses, which identify devices on the local network segment. Unlike IP addresses, MAC addresses are only used for communication within the local network. The EtherType field (0x0800) indicates that the payload carried inside the Ethernet frame is an IPv4 packet

IPv4 Header Analysis

```

> Internet Protocol Version 4, Src: 192.168.0.109, Dst: 34.223.124.45
  - 0100 .... = Version: 4
  - .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  - Total Length: 500
  - Identification: 0x056f (1391)
  > 010. .... = Flags: 0x2, Don't fragment
  - ...0 0000 0000 0000 = Fragment Offset: 0
  - Time to Live: 64
  - Protocol: TCP (6)
  - Header Checksum: 0xd373 [validation disabled]
  - [Header checksum status: Unverified]
  - Source Address: 192.168.0.109
  - Destination Address: 34.223.124.45
  - [Stream index: 15]

```

The IPv4 header is added by the Network Layer. Its main purpose is to provide logical addressing and routing information so that routers can forward packets across different networks.

The IPv4 header contains:

- Version: 4 (specifies that the used IPV4)
- Header Length: 20 bytes (indicate where the IP payload begins)
- Total Length: 500 (show the complete size of the IP Packet)

- Identification: 0x056f (1391) (used if the packet must be fragmented)
- Flags: 0x2 (control Fragmentation)
- Fragment Offset: 0
- Time to Live: (TTL): 64 (prevent Packets from looping forever)
- Protocol: TCP (6) (Identifies the next protocol)
- Header Checksum: 0xd373 (check corruption in the IP header)
- source IP Address: 192.168.0.109
- Destination IP Address: 34.223.124.45

The source IP identifies my computer, while the destination IP identifies the web server. The TTL value decreases each time the packet passes through a router, preventing packets from circulating indefinitely. The Protocol field contains the value 6, which indicates that the payload is a TCP segment.

TCP Header Analysis:

```

Transmission Control Protocol, Src Port: 50032, Dst Port: 80, Seq: 1, Ack: 1, Len: 448
- Source Port: 50032
- Destination Port: 80
- [Stream index: 49]
- [Stream Packet Number: 5]
> [Conversation completeness: Complete, WITH_DATA (31)]
- [TCP Segment Len: 448]
- Sequence Number: 1      (relative sequence number)
- Sequence Number (raw): 3056619266
- [Next Sequence Number: 449      (relative sequence number)]
- Acknowledgment Number: 1      (relative ack number)
- Acknowledgment number (raw): 3516608375
- 1000 .... = Header Length: 32 bytes (8)
> .... 0000 0001 1000 = Flags: 0x018 (PSH, ACK)
- Window: 63
- [Calculated window size: 64512]
- [Window size scaling factor: 1024]
- Checksum: 0x1fc6 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
> Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
> [Timestamps]
> [SEQ/ACK analysis]
- [Client Contiguous Streams: 1]
- [Server Contiguous Streams: 1]
- TCP payload (448 bytes)

```

The TCP header is added at the Transport Layer. TCP provides reliable communication by ensuring that packets are delivered correctly, in order, and without data loss.

The captured TCP segment contains:

- Source Port: 50032 (Represents the temporary port opened by my browser)
- Destination Port: 80 (port 80 indicates the request is being sent to an HTTP server)
- Sequence Number: 1 (Keeps data in the correct order)
- Acknowledgment Number: 1 (confirm previously received bytes)
- Header Length: 32 bytes (8)
- Flags: 0x018 (PSH,ACK) (PSH tells the receiver to process the data immediately. ACK acknowledges previous packets)
- Window Size: 1024 (indicates how much data can be received before acknowledgment)
- checksum: 0x1fc6 (verifies integrity of the TCP segment)

The destination port 80 indicates that the request is sent to an HTTP web server . TCP uses sequence and acknowledgment numbers to grantee reliable delivery. The checksum is used to detect transmission errors.

HTTP Header Analysis:

```
Hypertext Transfer Protocol
> GET /online HTTP/1.1\r\n
Host: lushclearglowingyawn.neverssl.com\r\n
Connection: keep-alive\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/149.0.0.0 Safari/537.36\r\n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8\r\n
Sec-GPC: 1\r\n
Accept-Language: en-US,en;q=0.7\r\n
Referer: http://neverssl.com/\r\n
Accept-Encoding: gzip, deflate\r\n
\r\n
[Response in frame: 638]
[Full request URI: http://lushclearglowingyawn.neverssl.com/online]
```

The HTTP protocol operates at the Application Layer.

The captured request is:

Get /online HTTP/1.1

GET (HTTP method requesting data)

"/online" (Requested resource)

HTTP/1.1 (protocol version)

This indicates that the client is requesting the "/online" resource from the web server.

The HTTP request also contains several important headers:

- Host
- User-agent
- Accept

- Accept-Encoding
- Accept-Language
- Connection

The Host header identifies the destination on website. The User-Agent identifies the browser used to send the request. The Accept headers specify the content types supported by the client, while the connection header indicates whether the TCP connection should remain open after the response.

Packet Bytes Mapping (manual byte-by-byte):

```

b4 0f 3b 64 9d 90 24 b2 b9 31 3f 09 08 00 45 00 ...;d.$ .1?...E.
01 f4 05 6f 40 00 40 06 d3 73 c0 a8 00 6d 22 df ...o@.@ .s...m".
7c 2d c3 70 00 50 b6 30 4f 02 d1 9b 2f 77 80 18 |.p.P.0 0.../w..
00 3f 1f c6 00 00 01 01 08 0a f4 cc 15 15 87 fb .?... ..
43 b2 47 45 54 20 2f 6f 6e 6c 69 6e 65 20 48 54 C.GET /o nline HT
54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6c 75 TP/1.1.. Host: lu
73 68 63 6c 65 61 72 67 6c 6f 77 69 6e 67 79 61 shclearg lowingya
77 6e 2e 6e 65 76 65 72 73 73 6c 2e 63 6f 6d 0d wn.never ssl.com.
0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 .Connect ion: kee
70 2d 61 6c 69 76 65 0d 0a 55 70 67 72 61 64 65 p-alive. Upgrade
2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73 -Insecur e-Reques
74 73 3a 20 31 0d 0a 55 73 65 72 2d 41 67 65 6e ts: 1..U ser-Agen
74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 t: Mozil la/5.0 (
58 31 31 3b 20 4c 69 6e 75 78 20 78 38 36 5f 36 X11; Lin ux x86_6
34 29 20 41 70 70 6c 65 57 65 62 4b 69 74 2f 35 4) Apple WebKit/5
33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c 20 6c 69 37.36 (K HTML, li
6b 65 20 47 65 63 6b 6f 29 20 43 68 72 6f 6d 65 ke Gecko ) Chrome
2f 31 34 39 2e 30 2e 30 2e 30 20 53 61 66 61 72 /149.0.0 .0 Safar
69 2f 35 33 37 2e 33 36 0d 0a 41 63 63 65 70 74 i/537.36 .Accept
3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61 70 70 6c : text/h tml,appl
69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d ication/ xhtml+xm
6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 6d l,application/xm
6c 3b 71 3d 30 2e 39 2c 69 6d 61 67 65 2f 61 76 l;q=0.9, image/av
69 66 2c 69 6d 61 67 65 2f 77 65 62 70 2c 69 6d if,image /webp,im
61 67 65 2f 61 70 6e 67 2c 2a 2f 2a 3b 71 3d 30 age/apng ,*/*;q=0
2e 38 0d 0a 53 65 63 2d 47 50 43 3a 20 31 0d 0a .8..Sec- GPC: 1..
41 63 63 65 70 74 2d 4c 61 6e 67 75 61 67 65 3a Accept-L anguage:
20 65 6e 2d 55 53 2c 65 6e 3b 71 3d 30 2e 37 0d en-US,en;q=0.7.
0a 52 65 66 65 72 65 72 3a 20 68 74 74 70 3a 2f .Referer : http:/
2f 6e 65 76 65 72 73 73 6c 2e 63 6f 6d 2f 0d 0a /neverss l.com/.
41 63 63 65 70 74 2d 45 6e 63 6f 64 69 6e 67 3a Accept-E ncoding:
20 67 7a 69 70 2c 20 64 65 66 6c 61 74 65 0d 0a gzip, d eflate.
0d 0a ..

```

Byte Offset	Hex Bytes	Meaning
0-5	b4 0f 3b 64 9d 90	Destination MAC Address
6-11	24 b2 b9 31 3f 09	Source MAC Address
12-13	08 00	EtherType = IPv4
14	45	Version (4) + IHL (5=20 Bytes)
15	00	DSCP/ECN
16-17	01 f4	Total Length
18-19	05 6f	Identification
20-21	40 00	Flags + Fragment Offset
22	40	TTL = 64
23	06	Protocol = TCP
24-25	d3 73	Header Checksum
26-29	c0 a8 00 6d	Source IP Address
30-33	22 df 7c 2d	Destination IP Address
34-35	c3 70	Source port
36-37	00 50	Destination Port
38-41	00 00 00 01	sequence Number
38-41	00 00 00 01	Acknowledgement Number
46	80	header Length
47	18	Flags
48-49	04 00	window Size
50-51	1f c6	checksum
52-53	00 00	Urgent Pointer

The Packet bytes shown in Wireshark represent the raw hexadecimal data transmitted over the network. Each group of bytes corresponds to a specific field in the Ethernet, IPv4, TCP or HTTP headers. By manually mapping these bytes to their respective headers fields, it is possible to understand exactly how a browser request is encapsulated before transmission.

HTTP GET Request Flow Chart:

```
Application Layer
|
HTTP GET Request
|
TCP Header Added
|
IPv4 Header Added
|
Ethernet Header Added
|
Physical Transmission
|
Destiantion
```

Process Memory Address Space:

The browser process (brave/chrome) occupies a virtual memory space provided by operating system.

During the HTTP GET request the process uses:

- Text Segment
Executable browser code
- data Segment
Global variables
- Heap
Stores dynamically allocated HTTP request objects, sockets buffer
- Stack
Stores function calls and local variables while constructing the HTTP request.

The browser itself does not directly access physical memory .It works within its own virtual address space, which is translated by the operating system's memory management unit (MMU)

Step 1;

When i typed a URL.

```
http://google.com
```

Browser Process in RAM.

RAM

```
+-----+
| Chrome Process |
+-----+
```

step 2:

Browser stores the URL temporary in the memory

```
google.com
```

this is stored in the temporary memory

step 3:

Browser calls the DNS resolver function to obtain the server's IP address

```
getaddrinfo("google.com")
```

Step 4:

Browser creates a TCP socket before establishing the connection

```
socket()
connect()
send()
recv()
```

Information of socket also remain in the memory

Step 5:

Browser creates the HTTP GET request.

```
GET / HTTP/1.1
Host: google.com
```

Step 6:

Operating System:

The operating system passes the HTTP request to the kernel's TCP/IP networking stack.

```
graph TD; Browser --> OS[Operating System]; OS --> TCP[TCP/IP Stack]; TCP --> NIC[NIC]; NIC --> Internet;
```

Browser
|
Operating System
|
TCP/IP Stack
|
NIC
|
Internet